TURING MACHINES

Turing Machine (TM) is modified version of the PDA and it is much more powerful than PDA. Instead of using stack as in PDA, the TM uses the tape to store the symbols.

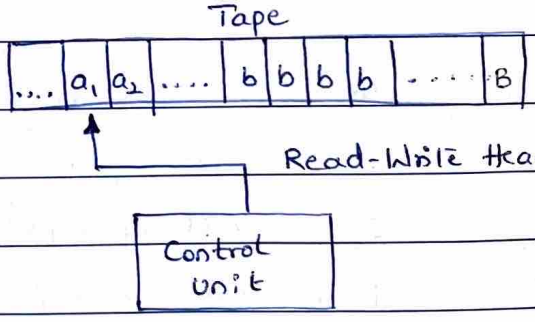
TM is a generalized machine which can recognize all types of languages,

Regular Languages generated from regular grammar known as type3 grammar
Context-free " " " Context free " " " type2 "
context-sensitive " " " Context sensitive " " " type 1 "
" " " type 0 grammar " " unrestricted "

Thus, TM can accept any language.

TM Model

It is a finite automaton connected to read-write head with the following components.



Tape is used to store the information and is divided into cells. Each cell can store the information of only one symbol. The string to be scanned will be stored from the leftmost position on the tape. The string should end with blanks. The tape is assumed to be infinite both on left side and right side of the string.

Read-Write Head can read a symbol from where it is pointing to and it can write into the tape to where it points to.

Control Unit: The reading from the tape or writing into the tape is determined by the control unit. The different moves performed by the

machine depends on the current scanned symbol and the current state. The control unit consults action table i.e., transition table and carry out the tasks.

The read-write head can move either towards left or right i.e., movement can be on both the directions. The various actions performed by the machine are

1. Change of state from one state to another state
2. The symbol pointing to by the read-write head can be replaced by another symbol.
3. The read-write head may move either towards left or towards right.

If there is no entry in the table for the current combination of symbol and state, then the machine will halt.

The TMs can be represented using various notations such as

- Transition tables
- Instantaneous descriptions
- Transition diagram

Def : The TM $M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

Q is set of finite states

Σ is set of input alphabets

T is set of tape symbols

δ is transition function from $Q \times T$ to $Q \times T \times \{L, R\}$

q_0 is the start state

B is a special symbol indicating blank characters

$F \subseteq Q$ is set of final states.

Def. ID : An ID of TM is a string in $\alpha q \beta$, where q is the current state, $\alpha\beta$ is the string made from tape symbols denoted by T, i.e., α and $\beta \in T^*$. The read-write head points to the first character of the substring β. The initial ID is denoted by $q\alpha\beta$ where q is the start state and the read-write head points to the first symbol of α from left. The final ID is denoted by $\alpha\beta q B$ where $q \in F$ is the final state and the read-write head points to the blank character denoted by B.

Def. Move : Let $M = (Q, \Sigma, T, \delta, q_0, B, F)$ be a TM. Let the ID of M be $a_1 a_2 a_3 \ldots a_{k-1} q a_k a_{k+1} \ldots a_n$

where $a_j \in T$ for $1 \le j \le n$, $q \in Q$ is the current state and a_k as the next symbol to be scanned.

If there is a transition,

$\delta(q, a_k) = (p, b, R)$ then the move of machine M will be

$a_1 a_2 a_3 \ldots a_{k-1} q a_k a_{k+1} \ldots a_n \vdash a_1 a_2 a_3 \ldots a_{k-1} b p a_{k+1} \ldots a_n$

If there is a transition,

$\delta(q, a_k) = (p, b, L)$

then the move of machine M will be

$a_1 a_2 a_3 \ldots a_{k-1} q a_k a_{k+1} \ldots a_n \vdash a_1 a_2 a_3 \ldots a_{k-2} p a_{k-1} b a_{k+1} \ldots a_n$

Language : Let $M = (Q, \Sigma, T, \delta, q_0, B, F)$ be a TM. The language L(M) accepted by M is defined as

$L(M) = \{w \mid q_0 w \vdash^* \alpha_1 p \alpha_2$ where $w \in \Sigma^*$, $p \in F$ and $\alpha_1, \alpha_2 \in T^*\}$

where $q_0 w$ is the initial ID and $\alpha_1 p \alpha_2$ is the final ID.

The set of all those words w in Σ^* which causes M to move from start state q_0 to the final state p.

BIT

Def : A language L is recursively enumerable, if it is accepted by a TM, i.e., given a string w which is input to TM, the machine halts and outputs Yes if it belongs to the language. If w does not belong to the language L, the TM halts and outputs NO.

The languages with Turing Machine which will always halts and output yes if it belongs to the language or output no if it does not belong to the language are called decidable languages or recursive languages. The TMs that always halt irrespective of whether they accept or not are a good model for an algorithm. If an algorithm exists to solve a given problem, then the problem is decidable otherwise it is un-decidable problem.

Q1. Obtain a TM to accept the language

$L = \{0^n 1^n \mid n \geq 1\}$

The TM to accept the language $L = \{0^n 1^n \mid n \geq 1\}$ is given by

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

$T = \{0, 1, X, Y, B\}$

$q_0 \in Q$ is the start state of machine

$B \in T$ is the blank symbol

$F = \{q_4\}$ is the final state

δ is shown below

① $\delta(q_0, 0) = (q_1, X, R)$ ✓

② $\delta(q_1, 0) = (q_1, 0, R)$ ✓

⑥ $\delta(q_1, Y) = (q_1, Y, R)$ ✓

③ $\delta(q_1, 1) = (q_2, Y, L)$ ✓

⑦ $\delta(q_2, Y) = (q_2, Y, L)$ ✓

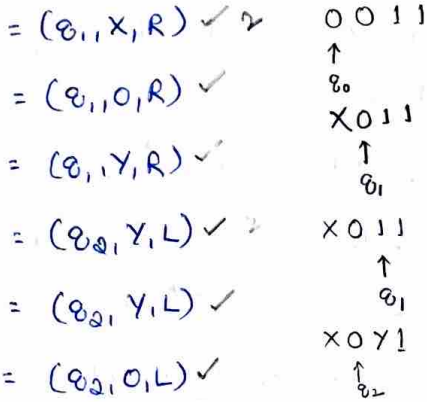
④ $\delta(q_2, 0) = (q_2, 0, L)$ ✓

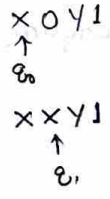
⑤ $\delta(q_2, X) = (q_0, X, R)$ ✓

⑧ $\delta(q_0, Y) = (q_3, Y, R)$

⑨ $\delta(q_3, Y) = (q_3, Y, R)$

⑩ $\delta(q_3, B) = (q_4, B, R)$

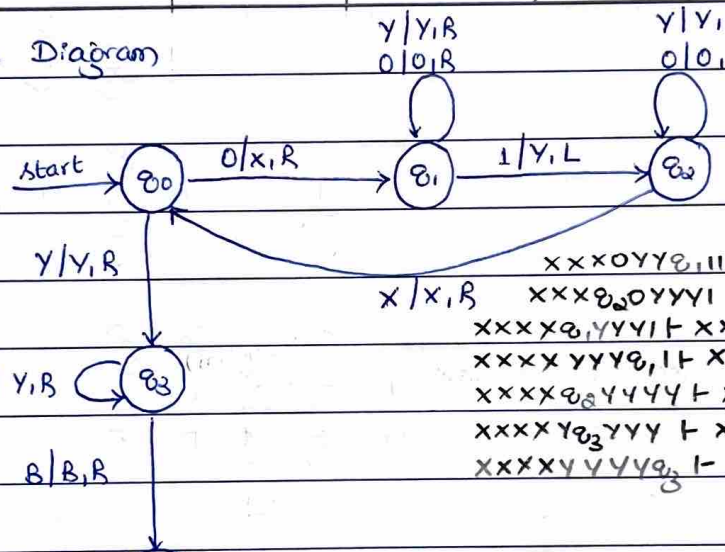




Transition Table

δ	Tape Symbols (T)				
States	0	1	X	Y	B
→ q_0	(q_1, X, R)	-	-	(q_3, Y, R)	-
q_1	$(q_1, 0, R)$	(q_2, Y, L)	-	(q_1, Y, R)	-
q_2	$(q_2, 0, L)$	-	(q_0, X, R)	(q_2, Y, L)	-
q_3	-	-	-	(q_3, Y, R)	(q_4, B, R)
* q_4	-	-	-	-	-

Transition Diagram



* Computation for 010

Initial ID = $q_0010 \vdash Xq_110 \vdash q_2XY0 \vdash Xq_0Y0 \vdash XYq_30$

$\delta(q_3, 0)$ is not defined, M halts. So, 010 is not accepted by M.

To accept the string: The sequence of moves or computations (IDs) for the string 0011 made by the TM is,

(Initial ID) $q_00011 \vdash Xq_111 \vdash X0q_11 \vdash Xq_20Y1 \vdash q_2X0Y1 \vdash$

$Xq_00Y1 \vdash XXq_1Y1 \vdash XXYq_11 \vdash XXq_2YY \vdash$

$Xq_2XYY \vdash XXq_0YY \vdash XXYq_3Y \vdash XXYYq_3 \vdash$

$XXYYBq_4$ (final ID)

Since the final state q_4 is reached, the string 0011 is accepted.

BIT

Initial ID
$q_00000111 \vdash$
$Xq_1000111 \vdash$
$X0q_100111 \vdash$
$X00q_10111 \vdash$
$X000q_1111 \vdash$
$X00q_20Y111 \vdash$
$X0q_200Y111 \vdash$
$Xq_20000Y111 \vdash$
$q_2X000Y111 \vdash$
$Xq_000Y111 \vdash$
$XXq_100Y111 \vdash$
$XX0q_10Y111 \vdash$
$XX00q_1Y111 \vdash$
$XX0q_20Y11 \vdash$
$XXq_200YY11 \vdash$
$Xq_2X00YY11 \vdash$
$XXq_000YY11 \vdash$
$XXXq_10YY11 \vdash$
$XXX0q_1YY11 \vdash$
$XXX0Yq_1Y11 \vdash$
$XXX0YYq_111 \vdash XXX0Yq_2YY1 \vdash XXX0q_20YY1 \vdash$
$XXXq_200YY1 \vdash XXq_2X0YY1 \vdash XXXq_00YY1 \vdash$
$XXXXq_1YY1 \vdash XXXXYq_1Y1 \vdash XXXXYYq_11 \vdash$
$XXXXYYYq_11 \vdash XXXXYYq_2YY \vdash XXXXq_2YYY \vdash$
$XXXXq_2YYYY \vdash XXXq_2XYYY \vdash XXXXq_0YYYY \vdash$
$XXXXYq_3YYY \vdash XXXXYYq_3YY \vdash XXXX YYYq_3Y \vdash$
$XXXXYYYYq_3 \vdash XXXXYYYBq_4$
(final ID)

Q2. Obtain a TM to accept the language

$$L(M) = \{0^n 1^n 2^n \mid n \geq 1\}$$

The TM to recognize the language $L = \{0^n 1^n 2^n \mid n \geq 1\}$ is given by

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

$\Sigma = \{0, 1, 2\}$

$T = \{0, 1, 2, X, Y, Z, B\}$

q_0 is the start state

B is blank character

$F = \{q_6\}$ is the final state

δ is shown below

1 $\delta(q_0, 0) = (q_1, X, R)$

13 $\delta(q_0, Y) = (q_4, Y, R)$

2 $\delta(q_1, 0) = (q_1, 0, R)$

3 $\delta(q_1, 1) = (q_2, Y, R)$

10 $\delta(q_1, Y) = (q_1, Y, R)$

4 $\delta(q_2, 1) = (q_2, 1, R)$

5 $\delta(q_2, 2) = (q_3, Z, L)$

11 $\delta(q_2, Z) = (q_2, Z, R)$

8 $\delta(q_3, 0) = (q_3, 0, L)$

6 $\delta(q_3, 1) = (q_3, 1, L)$

12 $\delta(q_3, Z) = (q_3, Z, L)$

7 $\delta(q_3, Y) = (q_3, Y, L)$

9 $\delta(q_3, X) = (q_0, X, R)$

15 $\delta(q_4, Z) = (q_5, Z, R)$

14 $\delta(q_4, Y) = (q_4, Y, R)$

16 $\delta(q_5, Z) = (q_5, Z, R)$

17 $\delta(q_5, B) = (q_6, B, R)$

$q_0 0 0 1 1 2 2$

$X q_1 0 1 1 2 2$

$X 0 q_1 1 1 2 2$

$X 0 Y 1 q_2 2 2$

$X 0 Y q_3 1 Z 2$

$X 0 q_3 Y 1 Z 2$

$X q_3 0 Y 1 Z 2$

$q_3 X 0 Y 1 Z 2$

$X q_0 0 Y 1 Z 2$

$X X q_1 Y 1 Z 2$

$X X Y q_1 1 Z 2$

$X X Y Y q_2 Z 2$

$X X Y Y Z q_2 2$

$X X Y Y q_3 Z Z$

$X X Y q_3 Y Z Z$

$X X q_3 Y Y Z Z$

$X X q_0 Y Y Z Z$

$X X Y q_4 Y Z Z$

$X X Y Y q_4 Z Z$

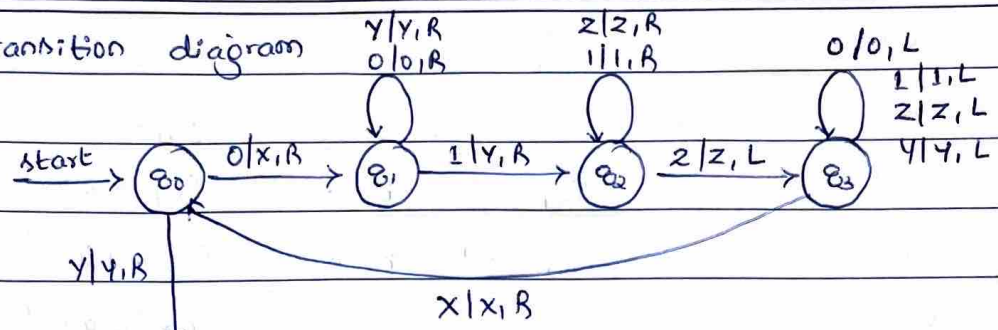
$X X Y Y Z q_5 Z$

$X X Y Y Z Z q_5$

Transition Table

δ / Tape Symbols (T)	0	1	2	X	Y	Z	B
$\rightarrow q_0$	(q_1, X, R)	–	–	–	(q_4, Y, R)	1	1
q_1	$(q_1, 0, R)$	(q_2, Y, R)	–	–	(q_1, Y, R)	1	1
q_2	–	$(q_2, 1, R)$	(q_3, Z, L)	–	–	(q_2, Z, R)	1
q_3	$(q_3, 0, L)$	$(q_3, 1, L)$	–	(q_0, X, R)	(q_3, Y, L)	(q_3, Z, L)	–
q_4	–	–	–	–	(q_4, Y, R)	(q_5, Z, R)	–
q_5	–	–	–	–	–	(q_5, Z, R)	(q_6, B, R)
$*q_6$	–	–	–	–	–	–	–

Transition diagram



NOTE : $L = \{0^n 1^n 2^n \mid n \geq 0\}$

if n starts with zero then include the transition from start state to the final state on Blank symbol.

i.e., $\delta(q_0, B) = (q_6, B, R)$

Q3. Obtain a Turing machine to accept a palindrome consisting of a's and b's of any length.

ε
a
b
aa
bb
aba
bab
abba

The TM to accept strings of palindromes over {a,b} is given by

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7\}$

$\Sigma = \{a, b\}$

$T = \{a, b, B\}$

q_0 is the start state

B is the blank character

$F = \{q_7\}$

δ is shown below

(Initial ID) $q_0Bbaab \vdash Bq_1baab \vdash$

$Bbq_5aab \vdash Baq_5ab \vdash Baaq_5b \vdash$

$Baabq_5 \vdash Baaq_6b \vdash Baq_4aB \vdash$

$Bq_4aaB \vdash q_4BaaB \vdash Bq_1aaB \vdash$

$BBq_2aB \vdash BBaq_2B \vdash BBq_3aB \vdash$

$Bq_4BBB \vdash BBq_1BB \vdash BBBq_7B$

(final ID)

1 $\delta(q_0, B) = (q_1, B, R)$

2 $\delta(q_1, a) = (q_2, B, R)$

3 $\begin{cases} \delta(q_2, a) = (q_2, a, R) \\ \delta(q_2, b) = (q_2, b, R) \end{cases}$

4 $\delta(q_2, B) = (q_3, B, L)$

5 $\delta(q_3, a) = (q_4, B, L)$

10 $\delta(q_3, B) = (q_7, B, R)$

7 $\delta(q_1, b) = (q_5, B, R)$

8 $\begin{cases} \delta(q_5, a) = (q_5, a, R) \\ \delta(q_5, b) = (q_5, b, R) \\ \delta(q_5, B) = (q_6, B, L) \end{cases}$

9 $\delta(q_6, b) = (q_4, B, L)$

$\delta(q_6, B) = (q_7, B, R)$ (bb)

$\delta(q_1, B) = (q_7, B, R)$ (aa)

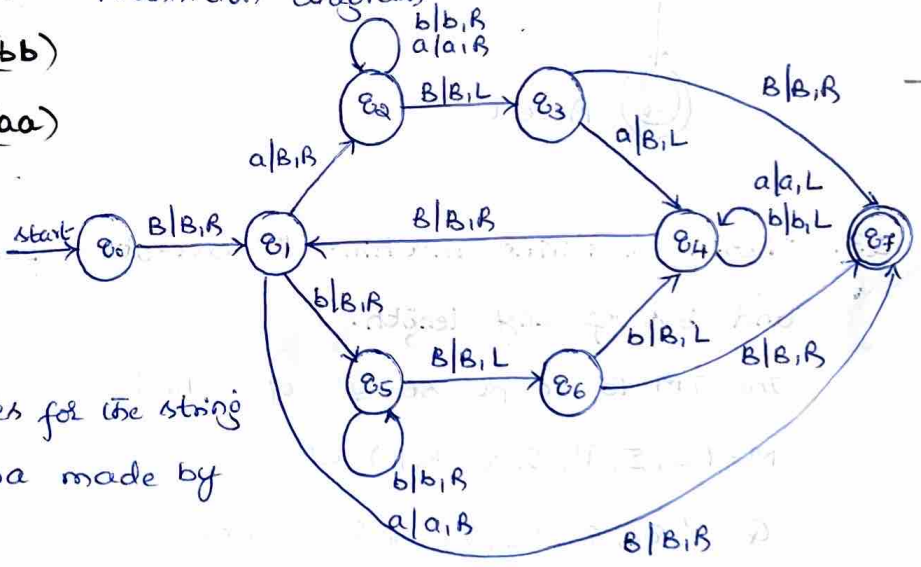
6 $\begin{cases} \delta(q_4, a) = (q_4, a, L) \\ \delta(q_4, b) = (q_4, b, L) \\ \delta(q_4, B) = (q_1, B, R) \end{cases}$

Transition Table

δ		Γ	
	a	b	B
$\rightarrow q_0$	-	-	(q_1, B, R)
q_1	(q_2, B, R)	(q_5, B, R)	(q_7, B, R)
q_2	(q_2, a, R)	(q_2, b, R)	(q_3, B, L)
q_3	(q_4, B, L)	-	(q_7, B, R)
q_4	(q_4, a, L)	(q_4, b, L)	(q_1, B, R)
q_5	(q_5, a, R)	(q_5, b, R)	(q_6, B, L)
q_6	-	(q_4, B, L)	(q_7, B, R)
$*\ q_7$	-	-	-

ababa

Transition diagram



The sequence of moves for the strings abba, aba and aaba made by the TM is,

(Initial ID) $q_0abba \vdash q_1abba \vdash Bq_2bba \vdash Bbq_2ba \vdash Bbbq_2a \vdash Bbbaq_2 \vdash Bbbq_3a \vdash Bbq_4bB \vdash Bq_4bbB \vdash q_4BbbB \vdash Bq_1bbB \vdash BBq_5bB \vdash BBbq_5B \vdash BBq_6bB \vdash Bq_4BBB \vdash BBq_1BB \vdash BBBq_7B$ (final ID) String accepted.

(Initial ID) $q_0Baba \vdash q_1aba \vdash Bq_2ba \vdash Bbq_2a \vdash Bbaq_2 \vdash Bbq_3a \vdash Bq_4bB \vdash q_4BbB \vdash Bq_1bB \vdash BBq_5B \vdash Bq_6BB \vdash BBq_7B$ String accepted. (final ID)

(Initial ID) $q_0Baaba \vdash q_1aaba \vdash Bq_2aba \vdash Baq_2ba \vdash Babq_2a \vdash Babaq_2 \vdash Babq_3a \vdash Baq_4bB \vdash Bq_4abB \vdash q_4BabB \vdash Bq_1abB \vdash BBq_2bB \vdash BBbq_2B \vdash BBq_3bB \vdash$ halts $\rightarrow$ String rejected

Q4. Obtain a TM to accept the language $L = \{ww^R \mid w \in (a+b)^*\}$ (OR).

The TM to accept the language $L = \{ww^R \mid w \in (a+b)^*\}$ is, Construct a TM that can accept the set of all even palindromes over $\{0,1\}$.

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7\}$

$\Sigma = \{a, b\}$

$T = \{a, b, B\}$

q_0 is the start state

B is the blank character

$F = \{q_7\}$ is the final state.

δ is shown below

1. $\delta(q_0, B) = (q_1, B, R)$
2. $\delta(q_1, a) = (q_2, B, R)$
3. $\delta(q_2, a) = (q_2, a, R)$
   $\delta(q_2, b) = (q_2, b, R)$
4. $\delta(q_2, B) = (q_3, B, L)$
5. $\delta(q_3, a) = (q_4, B, L)$
8. $\delta(q_1, b) = (q_5, B, R)$
9. $\delta(q_5, a) = (q_5, a, R)$
   $\delta(q_5, b) = (q_5, b, R)$
10. $\delta(q_5, B) = (q_6, B, L)$
11. $\delta(q_6, b) = (q_4, B, L)$
12. $\delta(q_1, B) = (q_7, B, R)$
6. $\delta(q_4, a) = (q_4, a, L)$
   $\delta(q_4, b) = (q_4, b, L)$
7. $\delta(q_4, B) = (q_1, B, R)$

(Initial ID) $q_0abaaba$

$\vdash Bq_2baaba$

$\vdash Bbq_2aaba$

$\vdash Bbaq_2aba$

$\vdash Bbaaq_2ba$

$\vdash Bbaabq_2a$

$\vdash Bbaaba q_2 -$

$\vdash Bbaabq_3a$

$\vdash Bbaaq_4bB$

$\vdash Bbaq_4abB$

$\vdash Bbq_4aabB$

$\vdash Bq_4baabB$

$\vdash q_4Bbaabb$

$\vdash Bq_1baabB$

$\vdash BBq_5aabB$

$\vdash BBaq_5abB \vdash BBaaq_5bB$

$\vdash BBaabq_5B \vdash BBaaq_6bB$

$\vdash BBaq_4aBB \vdash BBq_4aaBB$

$\vdash Bq_4BaaBB \vdash BBq_1aaBB$

$\vdash BBBq_2aBB \vdash BBBaq_2BB$

$\vdash BBBq_3aBB \vdash BBq_4BBBB$

$\vdash BBBq_1BBB \vdash BBBBq_7BB$

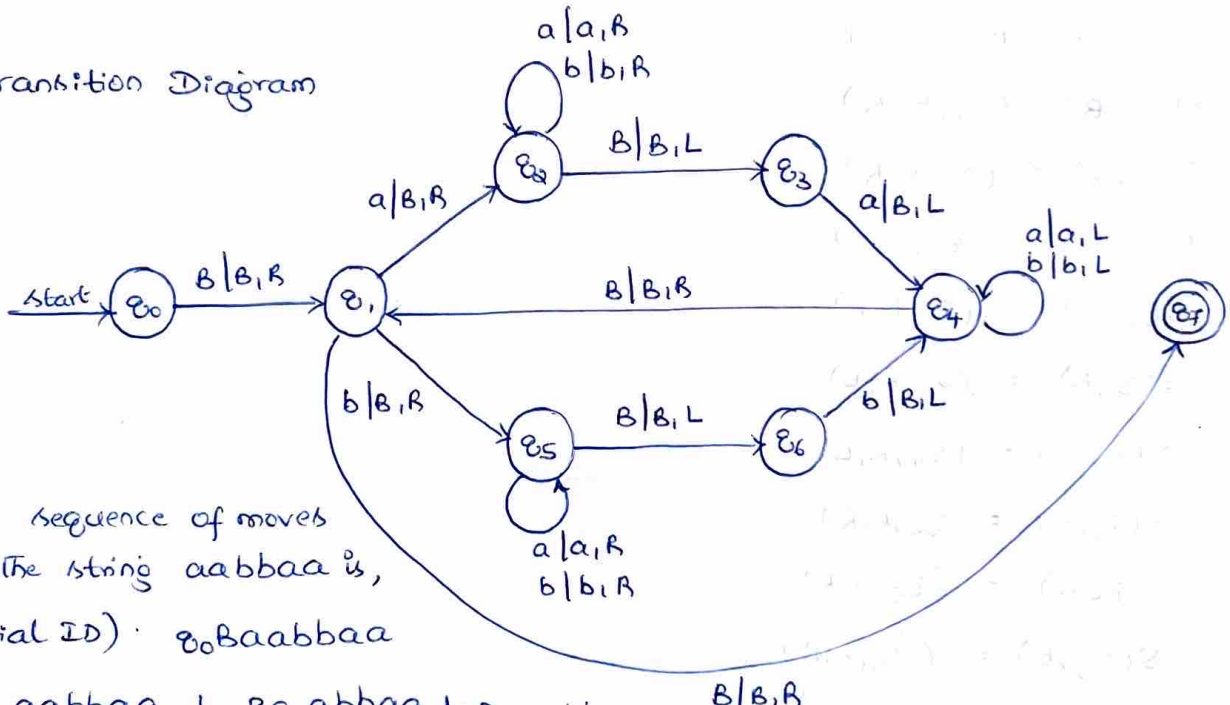
(final ID)

BIT

Transition Table

S	T		
	a	b	B
→q_0	–	–	(q_1, B, R)
q_1	(q_2, B, R)	(q_5, B, R)	(q_7, B, R)
q_2	(q_2, a, R)	(q_2, b, R)	(q_3, B, L)
q_3	(q_4, B, L)	–	–
q_4	(q_4, a, L)	(q_4, b, L)	(q_1, B, R)
q_5	(q_5, a, R)	(q_5, b, R)	(q_6, B, L)
q_6	–	(q_4, B, L)	–
*q_7	–	–	–

Transition Diagram



The sequence of moves for the string aabbaa is,

(Initial ID) $q_0Baabbaa$

$\vdash q_1aabbaa \vdash Bq_2abbaa \vdash Baq_2bbaa$

$\vdash Babq_2baa \vdash Babbq_2aa \vdash Babbaq_2a \vdash Babbaaq_2 \vdash Babbaq_3a \vdash$

$\vdash Babbq_4aB \vdash Babq_4baB \vdash Baq_4baB \vdash Bq_4abaB \vdash q_4BabbaB \vdash$

$\vdash Bq_1abbaB \vdash BBq_2bbaB \vdash BBbq_2baB \vdash BBbbq_2aB \vdash BBbbaq_2B \vdash$

$\vdash BBbbq_3aB \vdash BBbq_4baB \vdash BBq_4bbaB \vdash Bq_4BbbaB \vdash BBq_1bbaB \vdash BBBq_5baB$

$\vdash BBBbq_5aB \vdash BBBaq_5B \vdash BBBq_6aB \vdash BBq_4BBaB \vdash BBBq_1BaB \vdash BBBBq_7aB$

(final ID)

Q5. Obtain a TM to accept the language $L = \{0^n 1^n \mid n \geq 0\}$

The TM to accept the language $L = \{0^n 1^n \mid n \geq 0\}$ is given by

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{0, 1\}$

$T = \{0, 1, X, Y, B\}$

$q_0 \in Q$ is the start state of machine

$B \in T$ is the blank symbol

$F = \{q_4\}$ is the final state

δ is shown below

$\delta(q_0, 0) = (q_1, X, R)$

$\delta(q_0, Y) = (q_3, Y, R)$

$\delta(q_0, B) = (q_4, B, R)$

$\delta(q_1, 0) = (q_1, 0, R)$

$\delta(q_1, Y) = (q_1, Y, R)$

$\delta(q_1, 1) = (q_2, Y, L)$

$\delta(q_2, 0) = (q_2, 0, L)$

$\delta(q_2, Y) = (q_2, Y, L)$

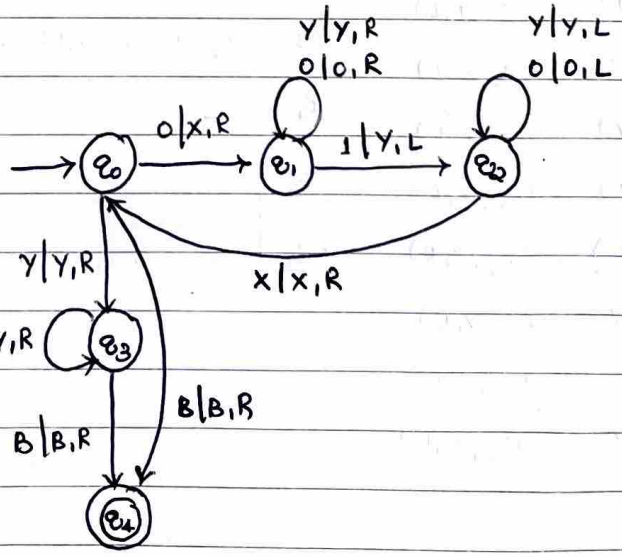
$\delta(q_2, X) = (q_0, X, R)$

$\delta(q_3, Y) = (q_3, Y, R)$

$\delta(q_3, B) = (q_4, B, R)$

Transition Table

δ	0	1	X	Y	B
$\rightarrow q_0$	(q_1, X, R)	–	–	(q_3, Y, R)	(q_4, B, R)
q_1	$(q_1, 0, R)$	(q_2, Y, L)	–	(q_1, Y, R)	–
q_2	$(q_2, 0, L)$	–	(q_0, X, R)	(q_2, Y, L)	–
q_3	–	–	–	(q_3, Y, R)	(q_4, B, R)
$*q_4$	–	–	–	–	–



Q6. Obtain a TM to accept the language $L(M) = \{0^n 1^n 2^n \mid n \geq 0\}$

The TM to recognize the language $L = \{0^n 1^n 2^n \mid n \geq 0\}$ is given by

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$

$\Sigma = \{0, 1, 2\}$

$T = \{0, 1, 2, X, Y, Z, B\}$

q_0 is the start state

B is blank character

$F = \{q_6\}$ is the final state

δ is shown below

$\delta(q_0, 0) = (q_1, X, R)$

$\delta(q_0, Y) = (q_4, Y, R)$

$\delta(q_0, B) = (q_6, B, R)$

$\delta(q_1, 0) = (q_1, 0, R)$

$\delta(q_1, 1) = (q_2, Y, R)$

$\delta(q_1, Y) = (q_1, Y, R)$

$\delta(q_2, 1) = (q_2, 1, R)$

$\delta(q_2, 2) = (q_3, Z, L)$

$\delta(q_2, Z) = (q_2, Z, R)$

$\delta(q_3, 0) = (q_3, 0, L)$

$\delta(q_3, 1) = (q_3, 1, L)$

$\delta(q_3, Z) = (q_3, Z, L)$

$\delta(q_3, Y) = (q_3, Y, L)$

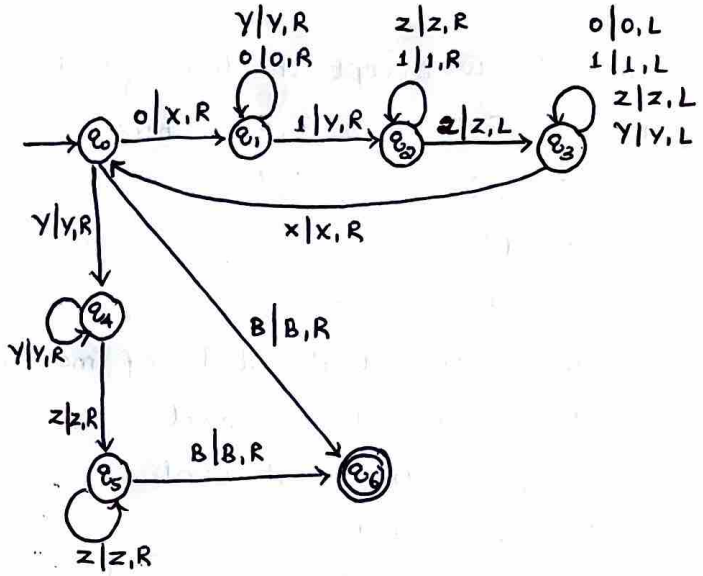
$\delta(q_3, X) = (q_0, X, R)$

$\delta(q_4, Z) = (q_5, Z, R)$

$\delta(q_4, Y) = (q_4, Y, R)$

$\delta(q_5, Z) = (q_5, Z, R)$

$\delta(q_5, B) = (q_6, B, R)$



Transition Table

δ	0	1	2	X	Y	Z	B
$\rightarrow q_0$	(q_1, X, R)	–	–	–	(q_4, Y, R)	–	(q_6, B, R)
q_1	$(q_1, 0, R)$	(q_2, Y, R)	–	–	(q_1, Y, R)	–	–
q_2	–	$(q_2, 1, R)$	(q_3, Z, L)	–	–	(q_2, Z, R)	–
q_3	$(q_3, 0, L)$	$(q_3, 1, L)$	–	(q_0, X, R)	(q_3, Y, L)	(q_3, Z, L)	–
q_4	–	–	–	–	(q_4, Y, R)	(q_5, Z, R)	–
q_5	–	–	–	–	–	(q_5, Z, R)	(q_6, B, R)
$*q_6$	–	–	–	–	–	–	–

Q7. Design a Turing Machine to accept even number of a's. Show the sequence of moves made by Turing Machine for the strings aa and aaa.

The TM to accept even number of a's is given by

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a\}$

$T = \{a, B\}$

$q_0 \in Q$ is the start state of machine

$B \in T$ is the blank symbol

$F = \{q_2\}$ is the final state

(i) δ is shown below

$\delta(q_0, B) = (q_2, B, R)$

$\delta(q_0, a) = (q_1, B, R)$

$\delta(q_1, a) = (q_0, B, R)$

(OR)

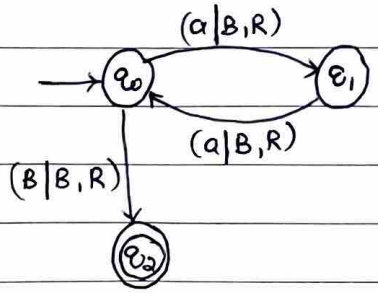
(ii) $\delta(q_0, B) = (q_2, B, R)$

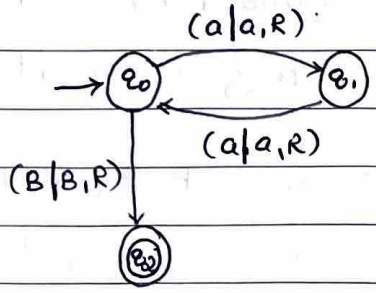
$\delta(q_0, a) = (q_1, a, R)$

$\delta(q_1, a) = (q_0, a, R)$

δ	a	B
$\rightarrow q_0$	(q_1, B, R)	(q_2, B, R)
q_1	(q_0, B, R)	-
$*q_2$	-	-

δ	a	B
$\rightarrow q_0$	(q_1, a, R)	(q_2, B, R)
q_1	(q_0, a, R)	-
$*q_2$	-	-





(i) The sequence of moves made by the TM for the string aa can be written as shown below

Initial ID = $q_0aaB \vdash Bq_1aB \vdash BBq_0B \vdash BBBq_2$

Since q_2 is the final state, the string aa is accepted by TM.

The sequence of moves made by the TM for the string aaa can be written as shown below

Initial ID = $q_0aaaB \vdash Bq_1aaB \vdash BBq_0aB \vdash BBBq_1B$

Since q_1 on B, no transition is defined, the given string aaa is rejected by TM.

(ii) The sequence of moves made by the TM for the string aa can be written as shown below

Initial ID = $q_0aaB \vdash aq_1aB \vdash aaq_0B \vdash BBBq_2$

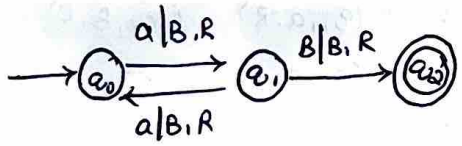
Since q_2 is the final state, the string aa is accepted by TM.

The sequence of moves made by the TM for the string aaa can be written as shown below

Initial ID = $q_0aaaB \vdash aq_1aaB \vdash aaq_0aB \vdash aaaq_1B$

Since q_1 on B, no transition is defined, the given string aaa is rejected by TM.

NOTE : odd number of a's



Initial ID = $q_0aaa \vdash Bq_1aa \vdash BBq_0a \vdash BBBq_1B \vdash BBBBq_2$

String is accepted by TM

Initial ID = $q_0aa \vdash Bq_1a \vdash BBq_0B$

String is rejected by TM

EXPERIMENT NO. ..... DATE .....

NAME OF EXPERIMENT ..... PAGE NO. 708

Q8. Design a Turing Machine to accept the strings that ends with 'b', over {a, b}.

The TM to accept the strings that ends with 'b' is given by

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$

$T = \{a, b, B\}$

$q_0 \in Q$ is the start state of machine

$B \in T$ is the blank symbol

$F = \{q_2\}$ is the final state

δ is shown below

$\delta(q_0, a) = (q_0, a, R)$

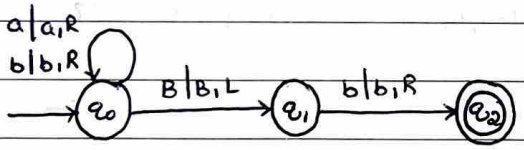
$\delta(q_0, b) = (q_0, b, R)$

$\delta(q_0, B) = (q_1, B, L)$

$\delta(q_1, b) = (q_2, b, R)$

Transition Table

δ	a	b	B
$\rightarrow q_0$	(q_0, a, R)	(q_0, b, R)	(q_1, B, L)
q_1	-	(q_2, b, R)	-
$*q_2$	-	-	-

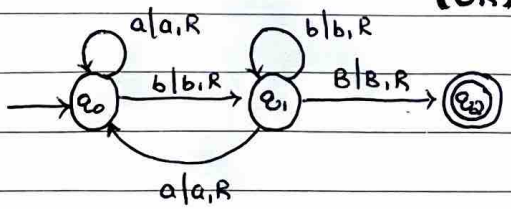


The sequence of moves made by the TM for the string abb can be written as shown below

Initial ID = $q_0abbB \vdash aq_0bbB \vdash abq_0bB \vdash abbq_0B \vdash abq_1bB \vdash abbq_2B$

Since q_2 is the final state, the string abb is accepted by TM.

(OR)



Initial ID = $q_0abbB \vdash aq_0bbB \vdash abq_1bB \vdash abbq_1B \vdash abbBq_2$

Q9. Construct a TM to accept the set L of all strings over {0,1} ending with 010.

The TM to accept the set of all strings ending with 010 is given by

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma = \{0, 1\}$

$T = \{0, 1, B\}$

$q_0 \in Q$ is the start state of machine

$B \in T$ is the blank symbol

$F = \{q_5\}$ is the final state

δ is shown below

$\delta(q_0, B) = (q_1, B, R)$

$\delta(q_1, 0) = (q_2, 0, R)$

$\delta(q_1, 1) = (q_1, 1, R)$

$\delta(q_2, 0) = (q_2, 0, R)$

$\delta(q_2, 1) = (q_3, 1, R)$

$\delta(q_3, 0) = (q_4, 0, R)$

$\delta(q_3, 1) = (q_1, 1, R)$

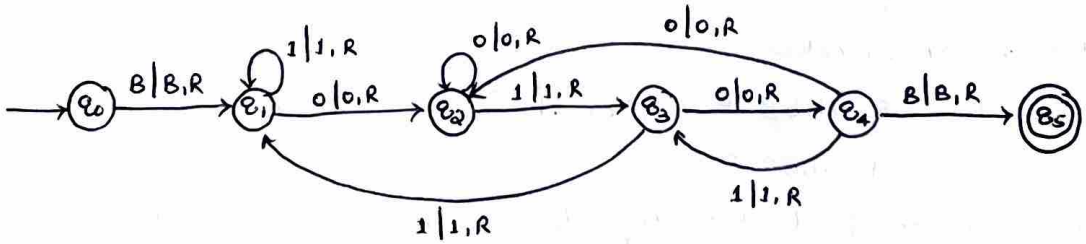
$\delta(q_4, 0) = (q_2, 0, R)$

$\delta(q_4, 1) = (q_3, 1, R)$

$\delta(q_4, B) = (q_5, B, R)$

Transition Table

δ	0	1	B
$\rightarrow q_0$	-	-	(q_1, B, R)
q_1	$(q_2, 0, R)$	$(q_1, 1, R)$	-
q_2	$(q_2, 0, R)$	$(q_3, 1, R)$	-
q_3	$(q_4, 0, R)$	$(q_1, 1, R)$	-
q_4	$(q_2, 0, R)$	$(q_3, 1, R)$	(q_5, B, R)
$*q_5$	-	-	-



Q 10. Design a Turing Machine to accept the strings that ends with 011 over $\{0,1\}$.

The TM to accept the set of all strings ending with 011 is given by

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma = \{0,1\}$

$T = \{0,1,B\}$

$q_0 \in Q$ is the start state of machine

$B \in T$ is the blank symbol

$F = \{q_5\}$ is the final state

δ is shown below

$\delta(q_0, B) = (q_1, B, R)$

$\delta(q_1, 0) = (q_2, 0, R)$

$\delta(q_1, 1) = (q_1, 1, R)$

$\delta(q_2, 0) = (q_2, 0, R)$

$\delta(q_2, 1) = (q_3, 1, R)$

$\delta(q_3, 0) = (q_2, 0, R)$

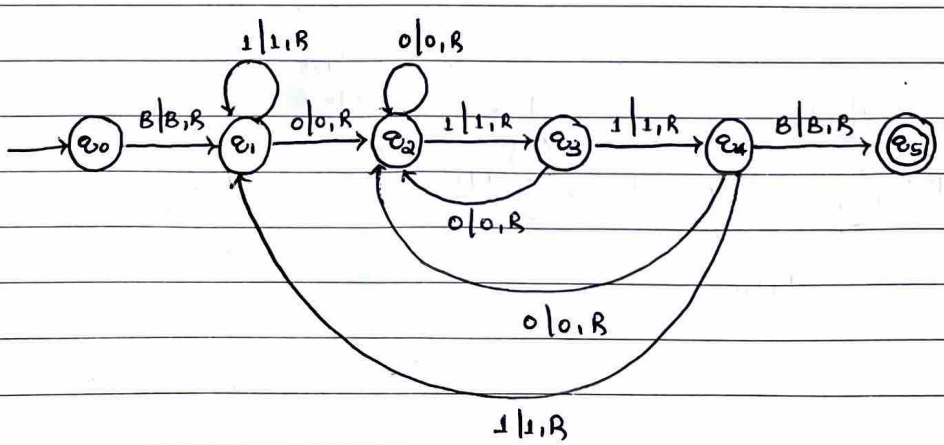
$\delta(q_3, 1) = (q_4, 1, R)$

$\delta(q_4, 0) = (q_2, 0, R)$

$\delta(q_4, 1) = (q_1, 1, R)$, $\delta(q_4, B) = (q_5, B, R)$

Transition Table

δ	0	1	B
$\rightarrow q_0$	-	-	(q_1, B, R)
q_1	$(q_2, 0, R)$	$(q_1, 1, R)$	-
q_2	$(q_2, 0, R)$	$(q_3, 1, R)$	-
q_3	$(q_2, 0, R)$	$(q_4, 1, R)$	-
q_4	$(q_2, 0, R)$	$(q_1, 1, R)$	(q_5, B, R)
$*q_5$	-	-	-



Q11. Construct a TM to accept the strings of 0's and 1's having the substring 001.

The TM to accept the strings of 0's and 1's having the substring 001 is given by

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma = \{0, 1\}$

$T = \{0, 1, B\}$

$q_0 \in Q$ is the start state of machine

$B \in T$ is the blank symbol

$F = \{q_5\}$ is the final state

δ is shown below

$\delta(q_0, B) = (q_1, B, R)$

$\delta(q_1, 0) = (q_2, 0, R)$

$\delta(q_1, 1) = (q_1, 1, R)$

$\delta(q_2, 0) = (q_3, 0, R)$

$\delta(q_2, 1) = (q_1, 1, R)$

$\delta(q_3, 0) = (q_3, 0, R)$

$\delta(q_3, 1) = (q_4, 1, R)$

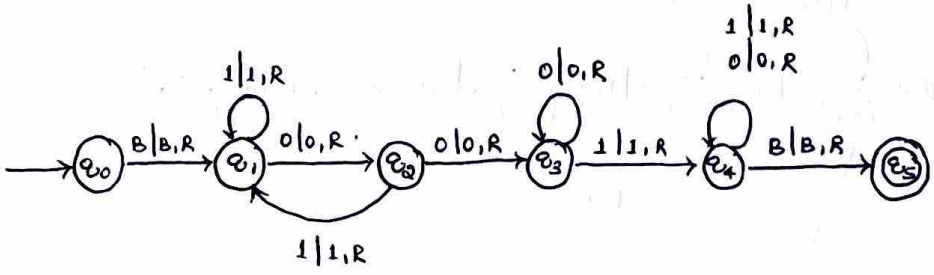
$\delta(q_4, 0) = (q_4, 0, R)$

$\delta(q_4, 1) = (q_4, 1, R)$

$\delta(q_4, B) = (q_5, B, R)$

Transition Table

δ	0	1	B
$\rightarrow q_0$	–	–	(q_1, B, R)
q_1	$(q_2, 0, R)$	$(q_1, 1, R)$	–
q_2	$(q_3, 0, R)$	$(q_1, 1, R)$	–
q_3	$(q_3, 0, R)$	$(q_4, 1, R)$	–
q_4	$(q_4, 0, R)$	$(q_4, 1, R)$	(q_5, B, R)
$*q_5$	–	–	–



EXPERIMENT NO. ..... DATE .....

NAME OF EXPERIMENT ..... PAGE NO. 710

Q12. Design a Turing Machine to recognize the language $\{1^n 2^n 3^n \mid n \geq 1\}$

The TM to recognize the language $\{1^n 2^n 3^n \mid n \geq 1\}$ is given by

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma = \{1, 2, 3\}$

$T = \{1, 2, 3, X, Y, Z, B\}$

$q_0 \in Q$ is the start state

$B \in T$ represents blank space

$F = \{q_5\}$ is the final state

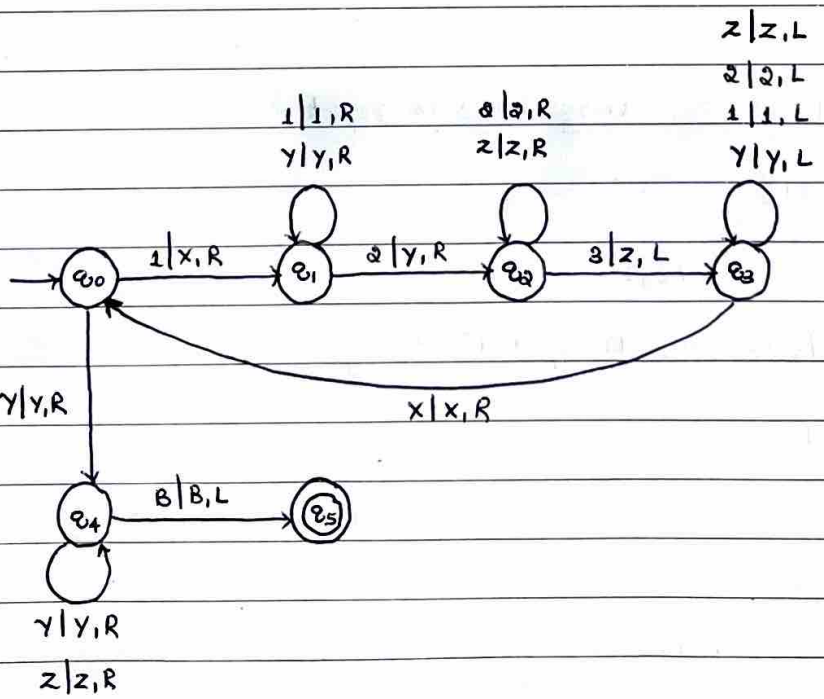
δ is shown below

NOTE: $L = \{1^n 2^n 3^n \mid n \geq 0\}$

If n value starts from zero then include a transition from start state to the final state on blanks symbol i.e., $\delta(q_0, B) = (q_5, B, R)$

Transition Table

δ	1	2	3	X	Y	Z	B
$\rightarrow q_0$	(q_1, X, R)	-	-	-	(q_4, Y, R)	-	-
q_1	(q_1, Y, R)	(q_3, Y, R)	-	-	(q_1, Y, R)	-	-
q_2	-	$(q_2, 2, R)$	(q_3, Z, L)	-	-	(q_2, Z, R)	-
q_3	$(q_3, 1, L)$	$(q_3, 2, L)$	-	(q_0, X, R)	(q_3, Y, L)	(q_3, Z, L)	-
q_4	-	-	-	-	(q_4, Y, R)	(q_4, Z, R)	(q_5, B, R)
$*q_5$	-	-	-	-	-	-	-



Computation for 112233

$q_0112233 \vdash Xq_1 12233 \vdash X1q_1 2233 \vdash X1Yq_2 233 \vdash X1Yq_2 33 \vdash X1Yq_3 Z3$
$\vdash X1q_3 YZ3 \vdash Xq_3 1YZ3 \vdash q_3 X1YZ3 \vdash Xq_0 1YZ3 \vdash XXq_1 YZ3 \vdash XXYq_1 Z3$
$\vdash XXYYq_2 Z3 \vdash XXYYZq_3 \vdash XXYYq_3 ZZ \vdash XXYq_3 YZZ \vdash XXq_3 YYZZ \vdash Xq_3 XYYZZ$
$\vdash XXq_0 YYZZ \vdash XXYq_4 YZZ \vdash XXYYq_4 ZZ \vdash XXYYZq_4 Z \vdash XXYYZZq_4 B \vdash XXYYZq_4 Z$

String is accepted

Computation for 1223

$q_01223 \vdash Xq_1 223 \vdash XYq_2 23 \vdash XY2q_2 3 \vdash XYq_3 2Z \vdash Xq_3 Y2Z \vdash q_3 XY2Z$
$\vdash Xq_0 Y2Z \vdash XYq_4 2Z$

$\delta(q_4, 2)$ is not defined. So, String 1223 is rejected.

Computation for 1123

$q_01123 \vdash Xq_1 123 \vdash X1q_1 23 \vdash X1Yq_2 3 \vdash X1q_3 YZ \vdash Xq_3 1YZ \vdash q_3 X1YZ$
$\vdash Xq_0 1YZ \vdash XXq_1 YZ \vdash XXYq_1 Z$

$\delta(q_1, z)$ is not defined. So, String is rejected.

Computation for 1233

$q_01233 \vdash Xq_1 233 \vdash XYq_2 33 \vdash Xq_3 YZ3 \vdash q_3 XYZ3 \vdash Xq_0 YZ3 \vdash XYq_4 Z3$
$\vdash XYZq_4 3$

$\delta(q_4, 3)$ is not defined. So, String 1233 is rejected.

Computation for 1213

$q_01213 \vdash Xq_1 213 \vdash XYq_2 13$

$\delta(q_2, 1)$ is not defined. So, String 1213 is rejected

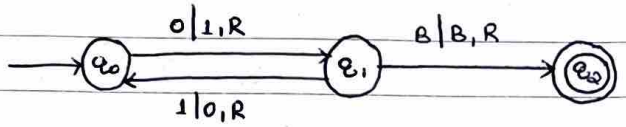
Computation for 2133

q_02133 $\delta(q_0, 2)$ is not defined. So, String 2133 is rejected.

Computation for 312

q_0312 $\delta(q_0, 3)$ is not defined. So, String 312 is rejected.

Q13. Let $M = (\{q_0, q_1, q_2\}, \{0,1\}, \{0,1,B\}, \delta, q_0, \{q_2\})$ where δ is given by $\delta(q_0,0) = (q_1,1,R)$, $\delta(q_1,1) = (q_0,0,R)$, $\delta(q_1,B) = (q_2,B,R)$, find $T(M)$.



$T(M) = \{(01)^n 0 \mid n \geq 0\}$

Q14. Construct a TM that accepts $L = \{0^{2^n} \mid n \geq 0\}$

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{0\}$

$T = \{0, X, B\}$

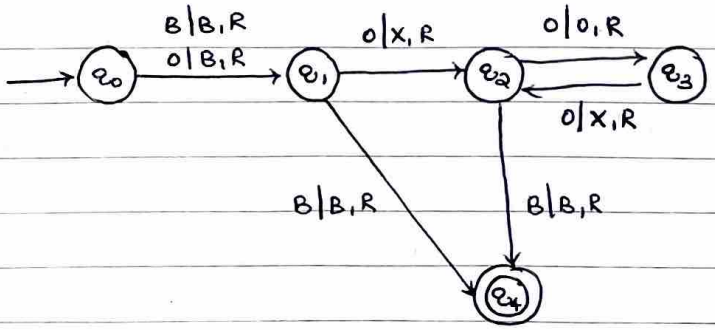
$q_0 \in Q$ is the start state

$B \in T$ represents blank symbol

$F = \{q_4\}$ is the final state

δ is shown below

δ	0	B
$\rightarrow q_0$	(q_1, B, R)	(q_1, B, R)
q_1	(q_2, X, R)	(q_4, B, R)
q_2	$(q_3, 0, R)$	(q_4, B, R)
q_3	(q_2, X, R)	-
$*q_4$	-	-



Q15. Design a TM that reads a string in $\{0,1\}^*$ and erases the rightmost symbol.

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{0, 1\}$

$T = \{0, 1, B\}$

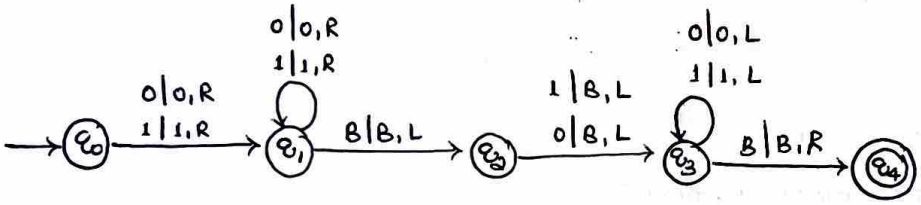
$q_0 \in Q$ is the start state

$B \in T$ represents blank symbol

$F = \{q_4\}$ is the final state

δ is shown below

δ	0	1	B
$\rightarrow q_0$	$(q_1, 0, R)$	$(q_1, 1, R)$	-
q_1	$(q_1, 0, R)$	$(q_1, 1, R)$	(q_2, B, L)
q_2	(q_3, B, L)	(q_3, B, L)	-
q_3	$(q_3, 0, L)$	$(q_3, 1, L)$	(q_4, B, R)
$*q_4$	-	-	-



Q16. Design a TM over $\{1, B\}$ which can compute a concatenation function over $\Sigma = \{1\}$. If a pair of words (ω_1, ω_2) is the input, the output has to be $\omega_1\omega_2$.

Let us assume that the two words ω_1 and ω_2 are written initially on the input tape separated by the symbol B.

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{1\}$

$T = \{1, B\}$

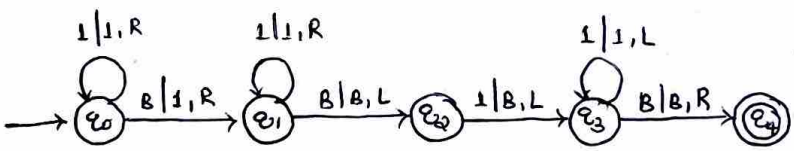
$q_0 \in Q$ is the start state

$B \in T$ represents blank symbol

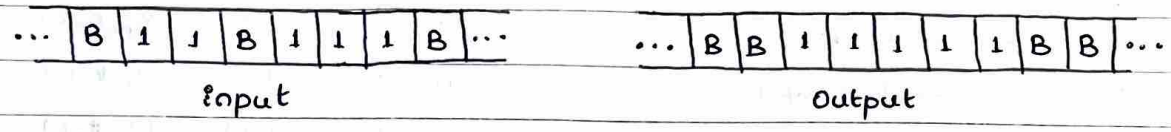
$F = \{q_4\}$ is the final state

δ is shown below

δ	1	B
$\rightarrow q_0$	$(q_0, 1, R)$	$(q_1, 1, R)$
q_1	$(q_1, 1, R)$	(q_2, B, L)
q_2	(q_3, B, L)	-
q_3	$(q_3, 1, L)$	(q_4, B, R)
$*q_4$	-	-



Example : if $\omega_1 = 11$ $\omega_2 = 111$



Computation for 11B111

$q_0 11B111 \vdash 1q_0 1B111 \vdash 11q_0 B111 \vdash 111q_1 111 \vdash 1111q_1 11 \vdash 11111q_1 1$
$\vdash 111111q_1 B \vdash 11111q_2 1B \vdash 1111q_3 1BB \vdash 111q_3 11BB \vdash 11q_3 111BB \vdash 1q_3 1111BB$
$\vdash q_3 11111BB \vdash q_3 B11111BB \vdash Bq_4 11111BB$ String is accepted.

Computation for 1B1

$q_0 1B1 \vdash 1q_0 B1 \vdash 11q_1 1 \vdash 111q_1 B \vdash 11q_2 1B \vdash 1q_3 1BB \vdash q_3 11BB$
$\vdash q_3 B11BB \vdash Bq_4 11BB$ String is accepted.

Q17. Design a TM for addition of two positive integers.

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{1\}$

$T = \{0, 1, B\}$

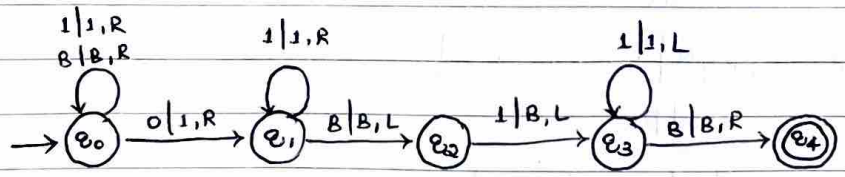
$q_0 \in Q$ is the start state

$B \in T$ is the blank symbol

$F = \{q_4\}$ is the final state

Transition Table

δ	0	1	B
$\rightarrow q_0$	$(q_1, 1, R)$	$(q_0, 1, R)$	(q_0, B, R)
q_1	-	$(q_1, 1, R)$	(q_1, B, L)
q_2	-	(q_3, B, L)	-
q_3	-	$(q_3, 1, L)$	(q_4, B, R)
$*q_4$	-	-	-



Example : ... B B 1 1 1 0 1 1 1 B B ...

input

(OR) ... B B 1 1 1 1 1 1 B B B ...

output

B.I.E.T

$M = (Q, \Sigma, T, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0\}$

$T = \{0, 1, B, \#\}$

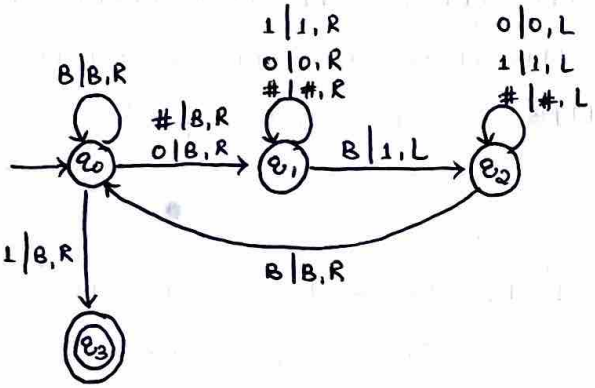
$q_0 \in Q$ is the start state

$B \in T$ is the blank symbol

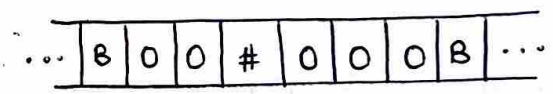
$F = \{q_3\}$ is the final state

Transition Table

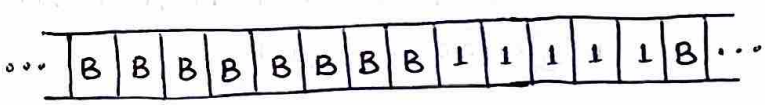
δ	0	1	#	B
$\rightarrow q_0$	(q_1, B, R)	(q_3, B, R)	(q_1, B, R)	(q_0, B, R)
q_1	$(q_1, 0, R)$	$(q_1, 1, R)$	$(q_1, \#, R)$	$(q_2, 1, L)$
q_2	$(q_2, 0, L)$	$(q_2, 1, L)$	$(q_2, \#, L)$	(q_0, B, R)
$*\ q_3$	–	–	–	–



Example :



input



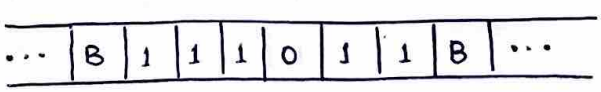
output

Q18. Design a TM that can compute proper subtraction, i.e., $m-n$, where m and n are positive integers. $m-n$ is defined as $m-n$ if $m > n$ and 0 if $m \leq n$.

Example : If $m = 3$, $n = 2$ $\quad m-n = 3 - 2 = 1$

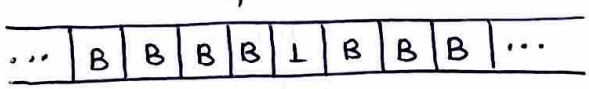
$= 111 - 11 = 1$

(i)



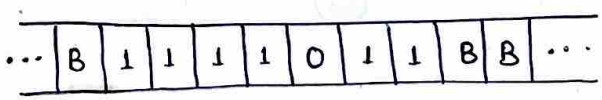
input

$3 - 2 = 1$



output

(ii)

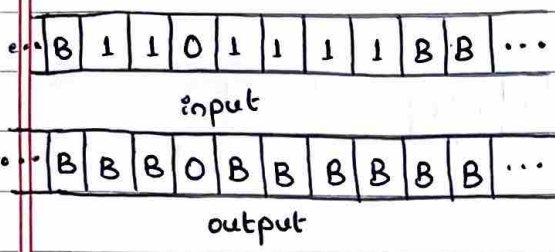


input

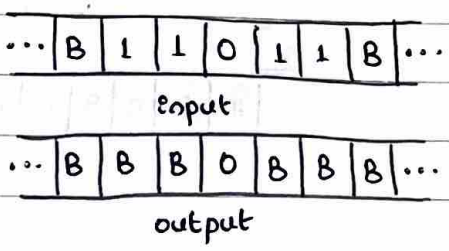
$4 - 2 = 2$

output

(iii) $2-4=0$



(iv) $2-2=0$



$M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$ where

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma = \{1\}$

$\Gamma = \{0, 1, B\}$

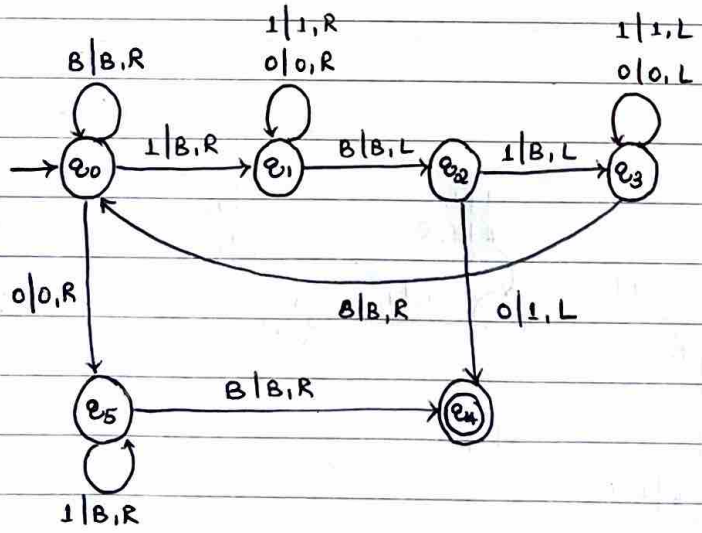
$q_0 \in Q$ is the start state

$B \in \Gamma$ is the blank symbol

$F = \{q_4\}$ is the final state

Transition Table

δ	0	1	B
$\rightarrow q_0$	$(q_5, 0, R)$	(q_1, B, R)	(q_0, B, R)
q_1	$(q_1, 0, R)$	$(q_1, 1, R)$	(q_2, B, L)
q_2	$(q_4, 1, L)$	(q_3, B, L)	-
q_3	$(q_3, 0, L)$	$(q_3, 1, L)$	(q_0, B, R)
$*q_4$	-	-	-
q_5	-	(q_5, B, R)	(q_4, B, R)

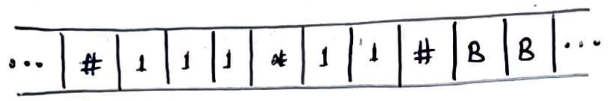


Computation for $3-2=1$ i.e., 111011

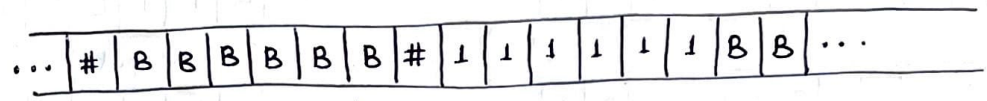
$q_0B111011B \vdash Bq_0111011B \vdash BBq_1110111B \vdash BB1q_110111B \vdash BB11q_10111B$
$\vdash BB110q_1111B \vdash BB1101q_111B \vdash BB11011q_1B \vdash BB1101q_2 1B \vdash BB110q_3 1BB$
$\vdash BB11q_3 01BB \vdash BB1q_3 101BB \vdash BBq_3 1101BB \vdash Bq_3B1101BB \vdash BBq_0 1101BB$
$\vdash BBBq_1 101BB \vdash BBB1q_1 01BB \vdash BBB10q_1 1BB \vdash BBB101q_1 BB \vdash BBB10q_2 1BB$
$\vdash BBB1q_3 0BBB \vdash BBBq_3 10BBB \vdash BBq_3B10BBB \vdash BBBq_0 10BBB \vdash BBBBq_1 0BBB$
$\vdash BBBB0q_1 BBB \vdash BBBBq_2 0BBB \vdash BBBq_4 B1BBB$

Q19. Design a TM for multiplication of two positive integers.

Example : 3 * 2 = 6



input



output

M = (Q, Σ, T, δ, q_0, B, F) where

Q = {$q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7$}

Σ = {1}

T = {#, B, 1, *, X}

$q_0 \in Q$ is the start state

$B \in T$ is the blank symbol

F = {q_7} is the final state

Transition Table

δ	#	B	1	*	X
→ q_0	(q_0, #, R)	–	(q_1, B, R)	(q_6, B, R)	–
q_1	–	–	(q_1, 1, R)	(q_2, *, R)	–
q_2	(q_5, #, L)	–	(q_3, X, R)	–	–
q_3	(q_3, #, R)	(q_4, 1, L)	(q_3, 1, R)	–	–
q_4	(q_4, #, L)	–	(q_4, 1, L)	–	(q_5, X, R)
q_5	–	(q_0, B, R)	(q_5, 1, L)	(q_5, *, L)	(q_5, 1, L)
q_6	(q_7, #, R)	–	(q_6, B, R)	–	–
*q_7	–	–	–	–	–